

CHAPTER 9

TROPICAL COMMUNITIES

CHAPTER OUTLINE AND SIGNIFICANT ELEMENTS

The bulk of this long chapter is taken up by a detailed discussion of CORAL REEFS. To facilitate this summary, I have grouped Nybakken’s discourse into eight subtopics. The second portion of this chapter covers MANGROVE FORESTS (MANGALS). Given the percentage of the world’s tropical coastlines populated by mangals (60–75%), these communities represent an important ecosystem.

Coral Reefs

In the shallow waters of the tropics, extensive regions are dominated by coral reefs. The single most important abiotic attribute of these regions is temperature. The location of the 20°C isotherm is a good rule of thumb in establishing the range of coral reefs.

I. Reef Distribution and Limiting Factors; Structure of Corals

20°C surface isotherm	emersion	salinity
ahermatypic corals	endosymbiotic algae	Scleractinia
Anthozoa	gastrodermis	sclerosepta
calcium carbonate	gastrovascular cavity	sea anemones
Cnidaria	hermatypic corals	sedimentation
cnidocytes (cells)	<i>Hydra</i>	septa
Coelenterata	light intensity	symbiosis
colonial vs. solitary	nematocysts (organelles)	turbidity
corallites	planula larva	upwelling
corals	polyps	zooxanthellae

1. Hermatypic corals, the reef-producing forms, possess zooxanthellae algae in their tissues. These algae aid the animal in the formation of the calcium carbonate matrix. This symbiosis is responsible for the binding of about 50% of the calcium that enters the world’s oceans (as CaCO₃), and, of course, this means that a considerable amount of CO₂ is deposited. As an ecosystem, coral reefs are among the most productive in the marine environment.
2. Throughout geologic history there have been reefs but the organisms that formed the reefs have changed through time. Currently, coral reefs form in waters with a mean minimum temperature of about 20°C. (Refer to Figure 9.1 in the textbook.) They also are limited to the upper, well-lit waters. Salinity, sedimentation rate, turbidity, wave action, and tidal extension are important to coral distribution as well.
3. Sea anemones are excellent models for a study of coral polyp anatomy. The only thing missing from those animals are the corallites or theca, with the sclerosepta partitions. Although composed of three layers, these animals are not triploblastic; they possess an epidermis and gastrodermis, and in between is a fibrous, jellylike layer possessing amoeboid cells. This layer is called the mesoglea. The presence of stinging nematocysts provides the etymon of this phylum: Cnidaria (G., *cnide*, nettle). Freshwater hydras are only adequate as models; they are very small and lack septa in the gastrovascular cavity. Some do, however, possess endosymbiotic algae in their tissues.

II. Types of Reefs; Origins of Reefs; Composition of Reefs

Coral reefs, like tropical rain forests, are among the most diverse and complex communities in the world. They are poorly understood and often overexploited.

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Alcyonacea	fire coral	<i>Millepora</i>
algal ridge	fringing reefs	patch reef
atolls	Gorgonacea	Scleratinia
barrier reefs	gorgonians	sea fans
Charles Darwin	Great Barrier Reef	sea whips
compensation theory	guyots	soft corals
coral-reef problem	<i>H.M.S. Beagle</i>	spicules
coralline algae	Hydrocorallina	stony corals
Eniwetok Atoll	lagoon reefs	subsidence theory

1. The coral-reef ecosystem is of interest for a variety of reasons: For example, coral reefs are important to the local economy for food and tourism. They also represent an ecological barometer of the health of tropical marine habitats and of the entire earth. (One could say that they function as the miner's canary.) In addition, they provide an opportunity to examine fundamental biological questions of species packing and zonation, symbiosis, coloniality, productivity, impact of catastrophe, etc. Some selected references follow:

Bellwood, D.R., and T.P. Hughes. 2001. Regional-scale assembly rules and biodiversity of coral reefs. *Science* 292: 1532.

Connolly, S.R., D.R. Bellwood, and T.P. Hughes. 2003. Indo-Pacific biodiversity of coral reefs: deviations from a mid-domain model. *Ecology* 84: 2178–2190.

Fraser, R.H., and D.J. Currie. 1996. The species richness–energy hypothesis in a system where historical factors are thought to prevail: coral reefs. *Am. Nat.* 148: 138–159.

Jackson, J.B.C. 1991. Adaptation and diversity of reef corals. *BioScience* 41: 475–482.

Karlson, R.H., and H.V. Cornell. 1999. Integration of local and regional perspectives on the species richness of coral assemblages. *Am. Zool.* 39: 104–112.

2. The structure of coral reefs varies, depending on the particular situation in which the reef develops, but three general categories can be recognized: atolls, barrier reefs, and fringing reefs. Reefs are very complex and rich communities, containing much more than just hermatypic corals.
3. The subsidence (compensation) theory, first developed by Charles Darwin, explains how coral atolls develop atop volcanic islands. However, it took more than 100 years for scientists to test this idea (and thus it became known as the coral-reef problem) by taking deep cores from atolls in the South Pacific.
4. Modern scleractinian reefs first appeared in the Mid-Triassic (ca. 235 m.y.a.). Earlier tropical reefs were dominated by other reef-forming organisms, including algae, sponges, clams (rudists), and corals that are now extinct. These other reef-forming organisms appear to have been subject to mass-extinction events, after which a new reef community evolved.

III. Coral Distribution and Reef Zonation

<i>Acropora</i>	leeward reef flat	seaward beach
algal ridge	leeward reef margin	small-scale patchiness
blowholes	leeward reef zone	spur and groove
buttress zone	Lithothamnion ridge	surge channels
lagoon beach	microatoll	windward reef margin
lagoon floor	outer seaward slope	zonation
lagoon reef	patchiness	
large-scale zonation	reef flat zone	

1. The Indo-Pacific region by far has more species of coral ($\approx 10\times$) than the Atlantic Ocean.
2. As noted in the last subsection, coral reefs are species rich, but the physical structure created by the reef community also is very complex. The textbook describes this complexity in detail and includes discussions of how wind, water, motion, light penetration, etc., act in the development of the reef.

IV. Atlantic and Indo-Pacific Reefs; Productivity

Coral reefs in the Atlantic and Indo-Pacific are very different, as described in the previous section, but these differences are not limited to the corals themselves. Sponges, mollusks, crustaceans, other cnidarians, and fish occur in different numbers in these two regions as well.

<i>Acanthaster planci</i>	elkhorn coral	oligotrophic
alcyonacean coral	gorgonians	platforms
anemone fish	hydrocoral	Pleistocene period
Atlantic reefs	Indo-Pacific reefs	POM
branch breaking	nocturnal corals	reef morphology
Caribbean reefs	nutrient cycling	sea fans
cyanobacteria	nutrient sink	sea whips
diurnal corals	octocorals	soft coral

1. Reef age and shape also differ. Pacific reefs are much older than Atlantic reefs; this difference is probably a function of the glacial periods during the Pleistocene period that affected Atlantic reefs more severely. Differences in zonation and life history of the coral populations of the Pacific and Atlantic reefs are evident, too.
2. Coral reefs are extraordinarily productive! Estimates of coral-reef productivity range from 1.5–5.0 kg C/m²/yr, or about 100× more productivity than that of open tropical oceans. This high level of productivity is probably due to the ability of this ecosystem to hold and cycle nutrients very efficiently.

V. Biology of Hermatypic Corals

This subsection is divided into three parts:

- Nutrition
- Growth and Calcification
- Sexual Maturity, Reproduction, and Recruitment

brain coral	larval recruitment	polyps
broadcast spawners	meroplankton	recruitment
brooders	nematocysts	synchronous spawn
calcification	nonsynchronous spawn	zooxanthellae
fragmentation	phototrophic	
host factor	planula larva	

1. While the plankton available to corals is abundant, it is insufficient to account for more than a fraction of the total C/E requirements of coral. Realization of this condition in the early 1970s led to much research in this area (e.g., Hinde, 1988; Sorokin, 1973).
 - a. Corals are predatory, using their nematocysts to capture planktonic organisms. About 10% of their food requirements are met this way. Corals also secrete mucus that can entrap small organisms, which are then shuttled by ciliary action into the gastrovascular cavity.
 - b. Zooxanthellae are symbiotic algae that live within the tissues of coral polyps. Their photosynthetic activities supply the animal with DOC, which greatly augments the energy budget of the coral. Free DOC from the surrounding waters also may be used by corals.
 - c. Finally, corals possess a common (interconnected) gastrovascular cavity. Thus, food materials may be transported from a region of the colony where they are in excess to a place of need.

Hinde, R. 1988. Symbiotic nutrition and nutrient limitation. In Choat, J.H., et al. (eds.). *Proceedings of the 6th international coral reef symposium*. Volume 1. Townsville, Australia: Symposium Executive Committee.

Sorokin, Y.I. 1973. On the feeding of some scleractinian corals with bacteria and dissolved organic matter. *Limnol. Oceanogr.* 18: 380–385.

2. Zooxanthellae are also important in calcification (that is, deposition of CaCO₃ in the coral skeletal matrix). Growth rates of corals are difficult to measure, partly because of their shape and partly

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because they grow slowly. Linear or better volumetric calculations of growth can be estimated for individual colonies. For larger areas, photographs taken from above a study plot can be compared from year to year (for example, see Figure 9.27 in the textbook). Thin sections of coral also can be examined and the growth lines studied much like tree rings. These slices also show the extent of the damage done by various organisms that bore into the skeletal matrix of hard corals. Use of radioisotopes permits the recording of rates of carbon fixation and the deposition of calcium carbonate.

Kizner, Z., R. Vago, and L. Vaky. 2001. Growth forms of hermatypic corals: stable states and noise-induced transitions. *Ecol. Model.* 141: 227–239.

Marshall, A.T. 1996. Calcification in hermatypic and ahermatypic corals. *Science* 271: 637–639.

- Both sexual and asexual reproduction are important to reef development. Asexual budding (binary fission) is important to the growth of individual colonies, and, in some cases, fragmentation begins new colonies. Sexual reproduction (and apparently some types of asexual reproduction) produces planula larvae that establish new colonies when they settle on a substrate.

VI. Species Interactions and Ecology of Reefs; Bacteria in Reef Systems; Role of Algae in Reef Systems

The first unit of this subsection is divided into five parts:

- Competition
- Predation
- Grazing
- Positive Interactions
- Regional Differences

aggressive pecking order
 ahermatypic corals
 algal gardens
 algal turf
 bioerosion
 coexistence
 commensalism
 competition
 competitive exclusion
 competitive hierarchies
 coralline algae
 corallivores
 cryptofauna
 dietary specialists

DOC
 El Niño
 encrusting
 exclusion experiments
 exploitative competition
 hermatypic corals
 histoincompatibility
 interference competition
 interspecific competition
 intraspecific competition
 mesenteric
 microbial loop
 mucous secretions
 mutualism

overgrowth
 overtopping
 POC
 scleractinian
 shade tolerant
 space occupiers
 space refuges
 sweeper polyps
 sweeper tentacles
 tissue necrosis
 waterborne chemicals
 weeding
 zooxanthellae

- Space is a commodity in very short supply within coral reefs. Corals and short macroalgae are the dominant occupiers of space in this ecosystem, and there is a fierce competition for this resource. Corals that grow fast are able to exploit the habitat more quickly than encrusting forms and thus can outcompete them by exploitative competition.
- Slower growing forms of coral are capable of engaging in combat with their competitors in a way similar to interclonal aggression in sea anemones. (See Chapter 6.) This type of competition is called interference competition. These behaviors and similar ones appear to be important in maintaining diversity in these highly competitive ecosystems.
- Some corals are shade tolerant, so they can grow deeper along the reef margin, where competition is less fierce. Corals must also compete with other organisms for space, especially algae, but also other encrusting invertebrates.
- Predation appears to be very important in structuring the coral reef community. Many predators graze on the coral itself, including gastropods, polychaetes, starfish (for example, *Acanthaster planci*), and fish. Corallivorous fish consume pieces of the coral skeleton, usually specializing on those

branching corals that grow quickly. This grazing influences the structure of the coral community. Undigested bits of coral skeletons deposited onto the reef as coralline sand and sediment by corallivores reduce reef structure, but the sediment becomes part of the larger reef matrix.

Benzie, J.A.H. 1999. Major genetic differences between crown-of-thorns starfish (*Acanthaster planci*) populations in the Indian and Pacific Oceans. *Evolution* 53: 1782–1795.

De'ath, G., and P.J. Moran. 1998a. Factors affecting the behaviour of crown-of-thorns starfish (*Acanthaster planci* L.) on the Great Barrier Reef: 1—patterns of activity. *J. Exp. Mar. Bio. Eco.* 220: 83–106.

De'ath, G., and P.J. Moran. 1998b. Factors affecting the behaviour of crown-of-thorns starfish (*Acanthaster planci* L.) on the Great Barrier Reef: 2—Feeding preferences. *J. Exp. Mar. Bio. Eco.* 220: 107–126.

5. Herbivorous fish also influence coral reefs. By removing algae—a major competitor with corals for space—herbivorous fish provide an opportunity for the coral to maintain itself. Algae fight back, so to speak, with their own defenses, including laying down calcium carbonate in their tissues and producing chemicals that are unpalatable to corals.
6. The role of bacteria in coral reefs is not well known. Production depends on a C/E source such as DOC. The numbers of bacteria present in this ecosystem probably represent a large source of energy that is available to the coral (Sorokin, 1973) and thus represent another link of the microbial loop (see Chapter 2).
7. Intense grazing pressure on algae by fish and invertebrates appears to be responsible for restricting this important space user. The catastrophic die-off of the grazing sea urchin, *Diadema*, in the Caribbean has demonstrated the importance of grazing in restraining algal growth, especially in regions where overfishing has reduced herbivorous fish. Algae provide other important roles in the reef as well. *Lithothamnion* cements smaller fragments of calcium carbonate into the larger matrix, while other algae bore into the coral skeletons.

VII. Ecology of Reef Fishes

camouflage	larval recruitment	reef flat
carrying capacity	lottery hypothesis	reef front
ciguatera	niche	rugosity
cleaning behavior	nocturnal	sense inversion
cleaning stations	nonequilibrium theory	specialization theory
competition model	postrecruitment competition	species richness
crinotoxin	predation-disturbance model	symbiosis
cuvierian organs	recruitment-limitation model	warning coloration
dinoflagellate	reef crest	

1. The same geographical variation (pattern) of species diversity of corals is reflected in coral-reef fishes. The highest diversities are found in the regions of the Philippines and Indonesia, with decreasing diversities heading out from this area. Species diversity of Atlantic reefs is also low.
2. The high species diversity of coral-reef fish reflects, in part, the great variety of habitats available in a reef complex. However, habitat diversity does not fully explain the large number of fish. Three ideas have been proposed to explain this phenomenon: the specialization theory (each fish species is highly specialized so that it is competitively superior in some place on the reef); the lottery hypothesis (the fish are not highly specialized, but produce large numbers of offspring, which by chance can occupy any unfilled space); and the nonequilibrium theory (unpredictable events of predation, weather, etc., reduce the various populations in random ways so that one species cannot eliminate its competitors).
3. Coral reef fishes seem to work two shifts: one diurnal and the other nocturnal. In some cases, the migration from nighttime feeding to daytime rest (for example, as exhibited by grunts) brings important nitrogen- and phosphorus-rich fecal material to the coral. The fish also have their preferences for specific coral species. (A reminder is in order about the use of such figures of speech as

“preference,” “selection,” and “choice.” We must be careful to let our students know that this simplistic language is steeped in teleological thought.)

4. Toxins appear to be a common theme in coral-reef organisms, in both the vertebrate and the invertebrate world.

VIII. Coral Reef Cryptofauna and Bioerosion; Succession and Stability in Coral Reefs; Catastrophic Mortality and Recovery of Reefs; Long-Term Dynamics

Given the relatively shorter length of human lives, coral reefs may seem to be stable ecosystems, but they are not. These ecosystems are dynamic and, according to historic geology, have experienced both good times and bad times.

<i>Acanthaster planci</i>	El Niño	runoff hypothesis
bioerosion	ENSO	Southern Oscillation
black-band disease	Great Barrier Reef	stability
catastrophe	hurricanes	succession
coral bleaching	opportunists	white-band disease
crown-of-thorns starfish	phase shift	
cryptofauna	predator-removal theory	

1. Many different invertebrates, including sponges, bryozoans, polychaetes, and tunicates, live in crevices of (opportunists) or burrow into (true borers) the calcium carbonate matrix of the coral reef. These organisms are collectively referred to as the cryptofauna because of their hidden lives. They have the ability to occupy space quickly and thus are at a competitive advantage relative to other forms. Some of these cryptic organisms cause erosion of the coral reef; together with that caused by noncryptic forms (for example, echinoids and grazing fish), the bioerosion caused by these organisms can be very significant.
2. Coral reefs appear to be stable features of the tropical marine environment. However, they do suffer from large-scale destructive forces such as large storms, population explosions of coral predators (*Acanthaster planci*), and regional climatic swings such as El Niño–Southern Oscillation (ENSO) and elevations of water temperature that appear to be correlated with coral bleaching. Disease-producing organisms also are serious problems for coral.
3. Human activities, of course, have caused significant damage to coral reefs. Loss of coral reefs significantly influences other populations, especially fishes.

Anon. 1991. Stupid diver tricks. *Sea Frontiers* 37(2): 18–19. [This reference is a list of coral-reef damage that divers can cause.]

Aronson, R.B., W.F. Precht, and I.G. Macintyre. 2000. Coral bleach-out in Belize. *Nature* 405: 36.

Buddemeir, R.W., and D.G. Fautin. 1993. Coral bleaching as an adaptive mechanism. *BioScience* 43(5): 320–326.

Connell, J.H., T.P. Hughes, and C.C. Wallace. 1997. A 30-year study of coral abundance, recruitment, and disturbance at several scales in space and time. *Ecol. Monogr.* 67: 461–488.

Normile, D. 2000. Some coral bouncing back from El Niño. *Science* 288: 941–942.

Mangrove Forests (Mangals)

Mangrove forests, or mangals, are composed of dense stands of flowering, terrestrial, halophytic trees and shrubs that live at the edge of the sea. Twelve genera of eight families make up mangals. This section of the text is less involved than the previous one, so I have not summarized it using subcategories. It is divided into six units:

- Structure and Adaptations
- Distribution
- Physical Conditions of Mangrove Forests
- Zonation

■ Associated Organisms
 ■ Succession and Mortality

black mangrove	mangrove forest	salt glands
climax stage	mangrove swamp	sedimentation
ecocatastrophe	marl	sere
facultative halophytes	mud skippers	succession
humus marl	peat	tidal forest
isopod	pioneer mangroves	tidal subsidy
levee	pneumatophores	vivipary
mangal	prop roots	white mangrove
mangrove crab	red mangrove	zonation
mangrove fauna	reverse osmosis	

1. Mangroves possess either shallow extensive roots that spread the sediment or distinctive prop roots that run from aboveground parts into the sediments. Shallow roots also send up air shafts called pneumatophores above the surface. Pneumatophores help conduct air into root tissues buried in anoxic sediments.

Salt, of course, is a constant problem for mangroves, and various adaptations have evolved to circumvent it. These adaptations include internal water storage, ability to tolerate a high salt concentration, salt-secreting glands, and other types of physiological adaptations that permit reverse osmosis in the roots.

Vivipary (continuous development of the embryo while still attached to the parent plant) and waterborne dispersal are common to many mangroves.

2. Mangroves enjoy a wider distribution than coral reefs; mangals develop on tropical and subtropical shores that are well protected from wave action. Small stands of mangroves may develop on less well protected shores or in slightly cooler climates, but they are subject to occasional destruction. For example, during the winter of 1985–1986, a severe frost destroyed many of the black-mangrove stands along the northwest coast of Florida.
3. Mangroves require habitats without significant wave action in order to become established, and by their growth, they further diminish wave action and increase sedimentation. Rates of sediment accumulation can be phenomenal, resulting in the accumulation of land at a rate as high as 200 m/yr!
4. No universal scheme describing zonation of mangals has been adapted, but a general pattern has been recognized. Two aspects of zonation should be kept in mind. One is horizontal: inland from the sea. The other is vertical: from the sediment upward into the branches of the mangroves. The influence of the tide on the distribution of organisms is manifest in both of these dimensions, and thus we can say that mangals are, in a sense, highly three dimensional.
5. The aboveground tissues provide a habitat for epiphytic colonists, including gastropods, barnacles, boring isopods, and hermit crabs. Some of these animals affect root growth negatively. Birds nesting in mangrove branches also can affect mangrove growth, but in this case positively, because their guano enriches the sediments. Various crustaceans and fishes use mangals as nurseries.
6. The classical view (Davis, 1940) argues that zonation patterns in mangals represent stages of successions or seres. Red mangroves are the pioneer sere, with other species making up the more advanced stages, eventually leading to a terrestrial environment. This view is not held by all workers. It appears that both biotic and abiotic factors are important in establishing zonation patterns. Storms can be significant factors contributing to massive mortality of mangroves, and once mangroves are lost, the locale can suffer significant erosion.

While storms and the boring isopod, *Sphaeroma terebrans*, can cause mortality, human abuse is by far the most important factor in the destruction of mangals. Use of herbicides, filling and dredging, and cutting, among other activities, are common mistreatments that we have imposed on these valuable ecosystems.

ENRICHMENT EXERCISES

Tropical Reefs

- Aronson, R.B., and W.F. Precht. 2000. Herbivory and algal dynamics on the coral reef at Discovery Bay, Jamaica. *Limno. Oceanogr.* 45: 251–255.
- Barnett, J., and W.N. Adger. 2003. Climate dangers and atoll countries. *Climatic Change* 61(3): 321–337.
- Briggs, J.C. 2004. Older species: a rejuvenation on coral reefs? *J. Biogeogr.* 31: 525–530.
- Goreau, T., and T. McClanahan. 2000. Conservation of coral reefs after the 1998 global bleaching event. *Cons. Biol.* 14: 5–15.
- Hawkins, J.P., and C.M. Roberts. 2004. Effects of fishing on sex-changing Caribbean parrotfishes. *Biol. Conser.* 115(2): 213–226.
- Lazereth, C.E., P. Willenz, and J. Navez. 2000. Sclerosponges as a new potential recorder of environmental changes: lead in *Ceratoporella nicholsoni*. *Geology* 28: 515–518.
- Sebens, K.P. 1994. Biodiversity of coral reefs: what are we losing and why? *Am. Zool.* 34: 115–133.
- Vernon, J.E.N. 1995. *Corals in space and time: the biogeography and evolution of the Scleractinia*. Ithaca, NY: Cornell University Press.

Mangals

For someone in Wisconsin to offer suggestions on exercises involving mangroves seems to me to be rather silly. Nevertheless, I do have some opportunities to see these interesting plants when I take my students on field trips to Florida. While we are in the field, I try to get the students to examine the plants and their surroundings. Some of the questions we have addressed include the following: How far do the pneumatophores extend up out of the sediment? Is there a pattern, and how might it be explained? Is there a relationship between the number of pneumatophores and the size of the plant between the area occupied by pneumatophores (surrounding the plant) and the size of the plant? Does the anoxic zone of the sediment extend to a uniform depth in the region where pneumatophores are present? (In this case, we just look for the depth at which the sediment turns to a dark color—an indication of the anoxic boundary.) How extensive are the prop roots? Is the sum of the cross-sectional area of the prop roots greater than, less than, or equal to the cross-sectional area of the main stem? What types of organisms make use of the mangrove as a habitat? If it has not rained recently, we wash off several leaves in a standard volume of distilled water and check the salinity. This exercise allows students to get a feel for salt secretion by the leaves. We also examine filling and erosion as a function of the number and distribution of mangrove plants. This analysis is especially instructive if there has been some recent die-off.

- Chen, R., and R.R. Twilley. 1998. A gap dynamic model of mangrove forest development along gradients of soil salinity and nutrient resources. *J. Eco.* 86: 37–51.
- Duke, N.C., J.A.H. Benzie, and J.A. Goodall. 1998. Genetic structure and evolution of species in the Mangrove genus *Avicennia* (Avicenniaceae) in the Indo-West Pacific. *Evolution* 52: 1612–1626.
- Ellison, A.M., E.J. Farnsworth, and R.R. Twilley. 1996. Facultative mutualism between red mangroves and root-fouling sponges in Belizean mangal. *Ecology* 77: 2431–2444.
- Lee, S.Y. 1999. Tropical mangrove ecology: Physical and biotic factors influencing ecosystem structure and function. *Austral. Eco.* 24: 355–366.
- Olafsson, E. 1998. Are wood-boring isopods a real threat to the well being of mangrove forests? *Ambio* 27: 760–761.
- Rützler, K., and Feller, I.C. 1996. Caribbean mangrove swamps. *Sci. Am.* 274(3): 94–99.

Fractals and Coral Reefs

A term project for the more energetic and mathematically minded student would be to compare the fine-scale fractal dimensions of different pieces of coral. I suggested the same sort of project in the chapter on rocky coasts (Chapter 9). Clearly, this information would be a silly curiosity unless it could be related to some

biological processes such as predator density or density of species using refugia. The whole idea here is, of course, to challenge the gifted student with a different way of thinking.

Barnsley, M.F. 1993. *Fractals everywhere*. Cambridge, MA: Academic Press Professional.

Pennycuik, C.J., and N.C. Kline. 1986. Units of measurement for fractal extent, applied to the coastal distribution of bald eagle nests in the Aleutian Islands, Alaska. *Oecologia* 68: 254–258.

Disease in Corals

Consultation of any good textbook in microbiology will provide you with a good overview of disease theory, including Koch's postulates. Also, you may wish to go to Brock (1998) to visit Robert Koch's seminal papers on disease. This information will help students understand just what it takes to determine the etiological agent of a specific disease.

Brock, T. 1998. *Milestones in microbiology*. Washington, DC: ASM Press.

